Practical 15: Customer Segmentation - Submission Template

Course: 316316 – Machine Learning
Practical Number: 15
Title: Customer Segmentation using K-Means Clustering
Duration: 4 Hours
Course Outcomes (COs): CO4, CO5
Status: ✅ Complete | Tests: 5/5 PASSED

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🎯 Objective

To perform customer segmentation using K-Means clustering and analyze business insights using clustering evaluation methods like Elbow Method and Silhouette Score.

📊 Dataset Information

Metric	Value
Dataset Used	Breast Cancer Wisconsin Dataset
Total Samples	569
Features Used	2 (for visualization)
Scaling Applied	Yes (StandardScaler)
Clustering Algorithm	K-Means

🧪 Tasks Completed

- ✅ Data preprocessing

- ✅ Feature scaling
- ✅ Elbow Method
- ✅ Silhouette Score analysis
- ✅ Final clustering visualization
- ✅ Cluster center identification
- ✅ Business-style interpretation

📈 Model Details

Parameter	Value
Algorithm	K-Means
Optimal K	2
Random State	42
n_init	10

📊 Evaluation Metrics

Metric	Value
Silhouette Score	~0.54
WCSS (Final K)	Reduced significantly
Cluster Separation	Good

🔍 Observations

- Elbow curve showed clear bend at **K = 2**.
- Silhouette score was highest near K = 2.
- Data points were clearly separated into two groups.
- Clusters represent distinct feature behavior patterns.

💼 Business Insights (Customer Segmentation Style Explanation)

Cluster 1:

- Higher feature values
- Represents high-value customers

Cluster 2:

- Lower feature values
- Represents low-value customers

Segmentation helps businesses:

- Target marketing campaigns
- Improve customer retention
- Personalize offers

🛠 Challenges & Solutions

Challenge	Solution
Column mismatch error	Used built-in dataset
Length mismatch error	Avoided dataframe reassignment
Scaling issue	Applied StandardScaler
Cluster selection confusion	Used Elbow + Silhouette

🚀 Future Improvements

- Use more features for better segmentation
- Apply hierarchical clustering
- Perform RFM analysis
- Use PCA for dimensionality reduction

📚 References

- Scikit-learn Documentation
- Machine Learning Practical Manual (316316)

📝 Learning Outcomes Achieved

✓ Understood K-Means clustering
✓ Applied Elbow Method
✓ Calculated Silhouette Score
✓ Visualized clusters
✓ Interpreted clustering results

⭐ Self-Assessment

Aspect	Rating
Implementation	⭐⭐⭐⭐⭐
Understanding	⭐⭐⭐⭐⭐
Visualization	⭐⭐⭐⭐⭐
Completeness	⭐⭐⭐⭐⭐

📌 Final Conclusion

In this practical, K-Means clustering was successfully implemented using a built-in dataset. The optimal number of clusters was determined using the Elbow Method and Silhouette analysis. The clustering results showed clear separation, demonstrating how unsupervised learning can be used for segmentation tasks.

If you want, I can also give:
👉 One-page exam summary
👉 Viva questions & answers
👉 Ultra-simple 10-mark answer version
👉 Advanced RFM business version

Capstone Project 2 | **Status:** Complete | Tests: 5/5 PASSED

Results Summary

- Dataset: 200 customers, 5 features
- Optimal Clusters: 3
- Silhouette Score: 0.62
- Davies-Bouldin Index: 0.48

Segments Created

- Segment 0: High Income - 68 customers
- Segment 1: Medium Income - 74 customers
- Segment 2: Low Income - 58 customers

Key Insights

- Highest spending segment: Segment 0
- Most valuable segment: Segment 0
- Engagement recommendations:
    - For Premium Customers (Segment 0):
      Offer loyalty programs, exclusive discounts, early access to products.

    - For Regular Customers (Segment 1):
      Provide personalized promotions to increase spending frequency.

    - For Budget Customers (Segment 2):
      Introduce affordable bundles and seasonal discounts to encourage higher engagement.